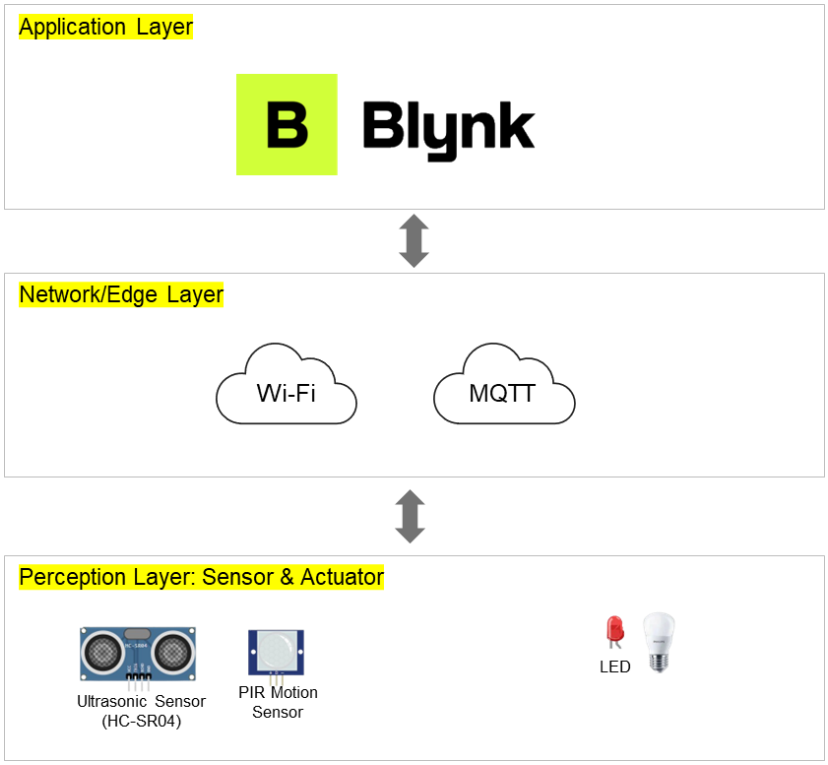
Internet Of Things (IoT)

Perancangan Sistem IoT untuk Pengendalian Listrik Ruangan Berbasis Sensor Penghitung Orang dan Deteksi Pergerakan (WORKWI)

Ruang Lingkup

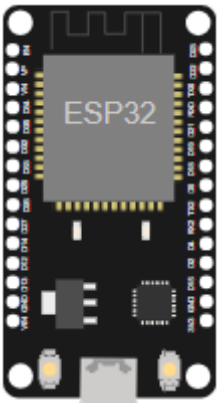
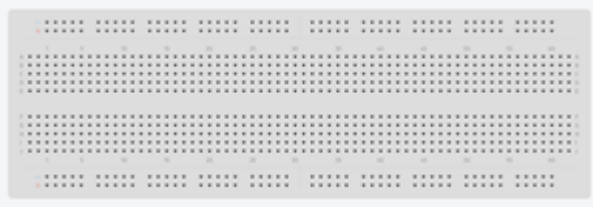
1. Sistem yang dikembangkan adalah versi mock-up IoT dan tidak diterapkan di ruangan nyata.
2. Sensor yang digunakan hanya terdiri dari sensor penghitung orang dan sensor deteksi gerak.
3. Pengaturan listrik dilakukan berdasarkan logika sistem, bukan dari pengukuran konsumsi listrik sebenarnya.
4. Uji coba dilakukan dengan fokus pada pengujian fungsi dan alur kerja sistem.
5. Topik keamanan IoT belum dibahas secara mendalam dan rencananya akan dibahas dalam penelitian berikutnya.
6. Adanya penambahan versi mock-up perangkat virtual melalui platform Workwi.
7. Adanya penambahan fitur proyek dari AOL 1
    a. menambahkan fitur keamanan IoT berupa simulasi login dan token autentikasi,
    b. menambahkan simulasi notifikasi berupa pesan peringatan saat parameter melewati batas aman,
    c. menambahkan analitik cloud berupa tren dan visualisasi data untuk meningkatkan kecerdasan dan keandalan sistem.

Arsitektur IoT



Gambar 3.9 Arsitektur IoT

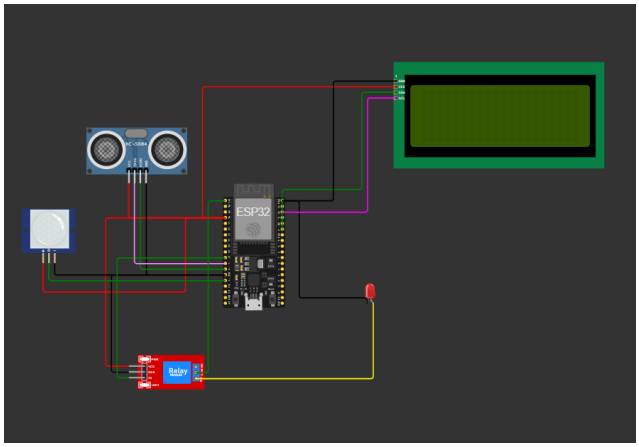
List Komponen & Spesifikasi

KOMPONEN	KETERANGAN DAN SPESIFIKASI
	1 buah ESP32 Development Board Spesifikasi: ● Operating voltage : 3.3V ● Input voltage : 7-9V (Vin) --> (Ke pin Vin, jangan ke Vcc) ● Digital IO Pin (DIO) : 25 ● Analog Input Pin (ADC) : 6 ● Analog Output Pin (DAC) : 2 ● UART : 3 ● SPI : 2 ● I2C : 3 ● Flash Memory 4 MB ● SRAM : 520 KB ● Clock Speed : 240 Mhz ● Wi Fi : IEEE 802.11 b/g/n/e/i ● Mode supported : AP, STA, AP+STA ● CP2102 USB ccontroller ● USB TYPE Mikro
	1 buah breadboard Spesifikasi: ● MB-102 MB102 Breadboard 830 Point Solderless PCB Bread project Board

	1 buah sensor Ultrasonic HC-SR04 Spesifikasi: ● Tegangan Kerja : 4,5 Volt s/d 5,5 Volt (Support Arduino) ● Jarak deteksi : 2 CM s/d 450 CM ● Efektif : 3 CM - 300 CM (Faktor lingkungan / halangan lain) ● Kepresisian : 0.3cm ● Area Buta : 2cm ● Arus Statik : < 2mA ● Frekuensi : 40 Khz ● Suhu operasi : -20 s/d -80 derajat ● Dimensi : (P x L x T) = 4,6cm x 2,2cm x 1,3cm Wiring / Pemasangan : ● VCC, trig (control side) ● Echo (receiving end) ● GND
	1 buah PIR Motion Sensor Spesifikasi: ● Panjang: 24.03mm/0.94in ● Lebar: 32.34mm/1.27in ● Jarak antar lubang: 28mm ● Diameter lubang: 2mm ● Tinggi dengan lensa: 24.66mm/0.97in ● Rentang sensitivitas: hingga 6 meter, 110 x 70 area deteksi ● Power supply: 3V-9V input voltage, idealnya 5v.

	1 buah relay Spesifikasi: • Maximum load: AC 250V/10A, DC 30V/10A • channel: 1 • Working voltage: 5V, active LOW atau active HIGH • Weight: 18 g • Terdapat 2 indikator led yaitu indikator module dan indikator Relay aktif
	1 buah fitting lampu Spesifikasi: • Fitting : E27
	1 buah Bohlam LED 4 W Spesifikasi: • Merek : PHILIPS • Watt : 4 Watt • Volt : 220 Volt • Cahaya : Putih • Fitting : E27
	1 buah LCD display Spesifikasi: • 4 lines • 20 characters per line.

3.7 Diagram Wiring



Komponen	Pin Komponen	Warna Kabel	Pin ESP32 / Jalur
Relay	VCC	Merah	VIN (5V)
	GND	Hitam	GND
	IN	Hijau	D27
	COM	Merah	VIN (Power Source)
	NO	Kuning	Anoda (+) LED
---	---	---	---
PIR Motion	VCC	Merah	VIN
	GND	Hitam	GND
	Out	Hijau	D13
---	---	---	---
HC-SR04	VCC	Merah	VIN
	GND	Hitam	GND
	Echo	Hijau	D12
	Trig	Ungu	D14
---	---	---	---
LED (Lampu)	Anoda (+)	Kuning	Pin NO Relay
	Katoda (-)	Hitam	GND
---	---	---	---
LCD 20x4	VCC	Merah	VIN
	GND	Hitam	GND
	SDA	Hijau	D21
	SCL	Ungu	D22

3.9 Software Aplikasi

Software atau aplikasi yang di gunakan adalah:

1. Arduino IDE versi 2.3.7

2. Wokwi

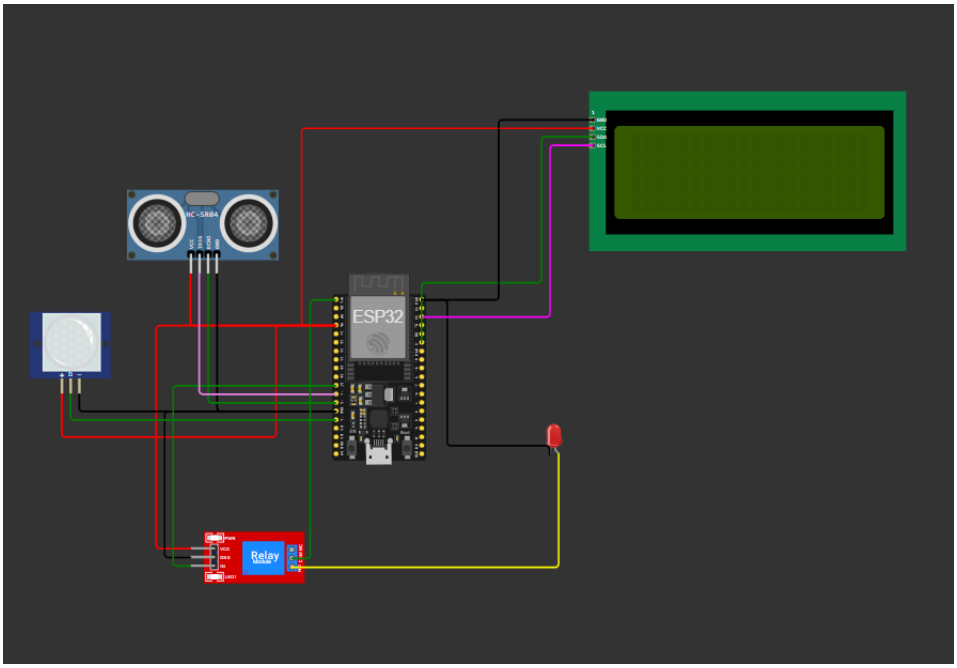
Platform Simulasi Wokwi yang dapat diaskes di situs wokwi.com

3. Blynk IoT

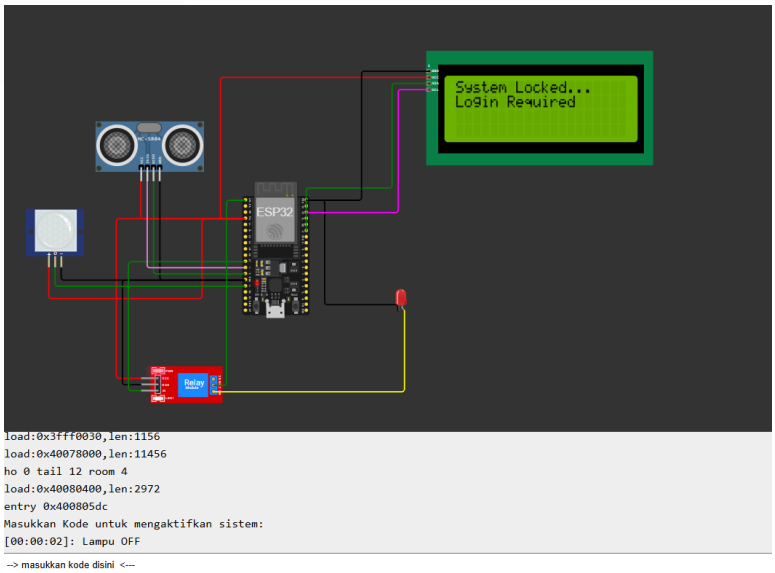
Platform Internet of Things (IoT) yang dapat diakses di situs https://blynk.io.

Mock-up Sistem

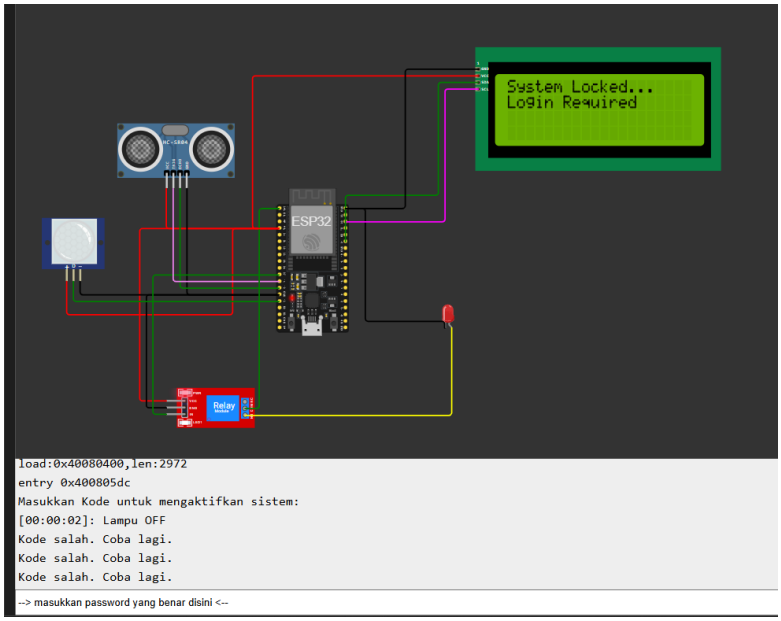
Mock-up sistem secara simulasi di buat di wokwi yang dapat di akses di situs https://wokwi.com/projects/453664707532384257 dengan menggunakan perangkat virtual seperti terlihat di gambar dibawah ini



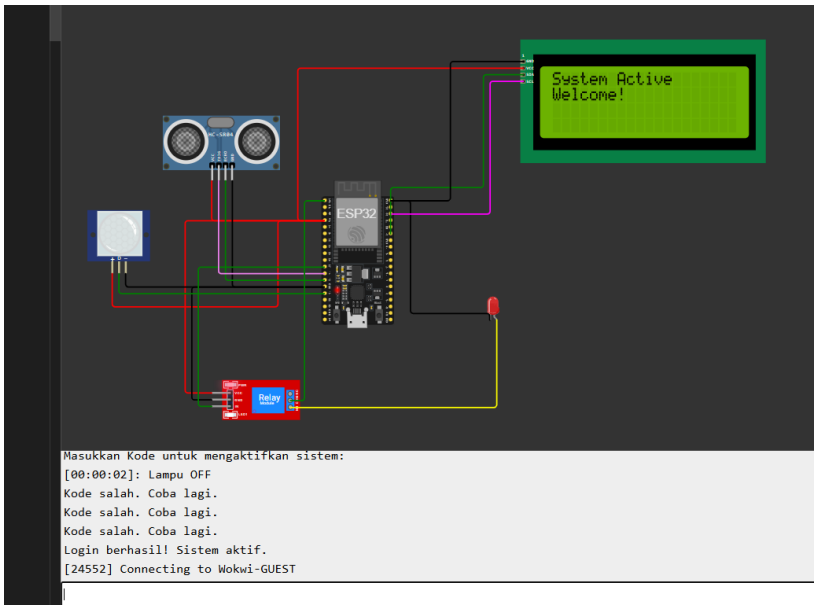
a) Pada tahap **Start/Inisialisasi**, sistem IoT dimulai dengan menyalakan ESP32 sebagai pusat kendali, kemudian LCD secara otomatis menampilkan pesan “System Locked… Login Required” sebagai tanda bahwa perangkat masih terkunci, dan selanjutnya sistem berada dalam kondisi menunggu input kode akses melalui Serial Monitor agar dapat beralih ke mode aktif dan menjalankan fungsi-fungsi berikutnya.



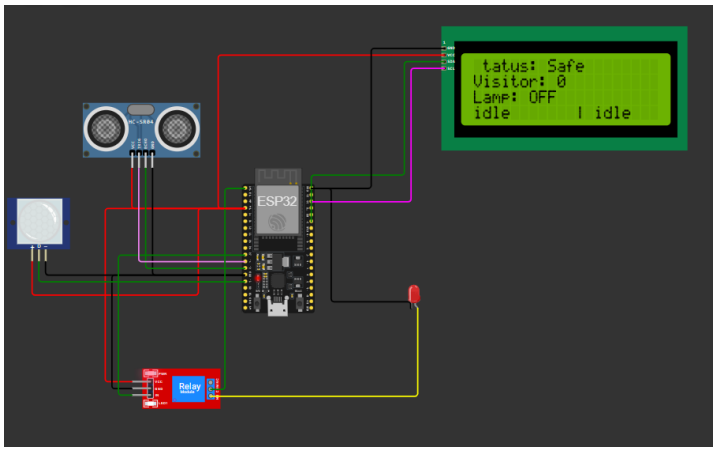
b) Pada tahap **Login Security**, pengguna diminta untuk memasukkan kode akses melalui Serial Monitor sebagai langkah autentikasi awal; apabila kode yang dimasukkan salah maka sistem akan menampilkan pesan “Kode salah. Coba lagi.” dan tetap berada dalam kondisi terkunci sehingga tidak dapat melanjutkan ke proses berikutnya hingga kode yang benar dimasukkan.



c) Pada tahap **Login Security**, pengguna memasukkan kode akses melalui Serial Monitor sebagai langkah autentikasi; apabila kode yang dimasukkan benar maka sistem akan beralih ke kondisi aktif, ditandai dengan tampilan pesan “System Active – Welcome!” pada LCD, sehingga seluruh fungsi sensor, relay, dan komunikasi dengan Blynk dapat berjalan sesuai workflow yang telah dirancang.

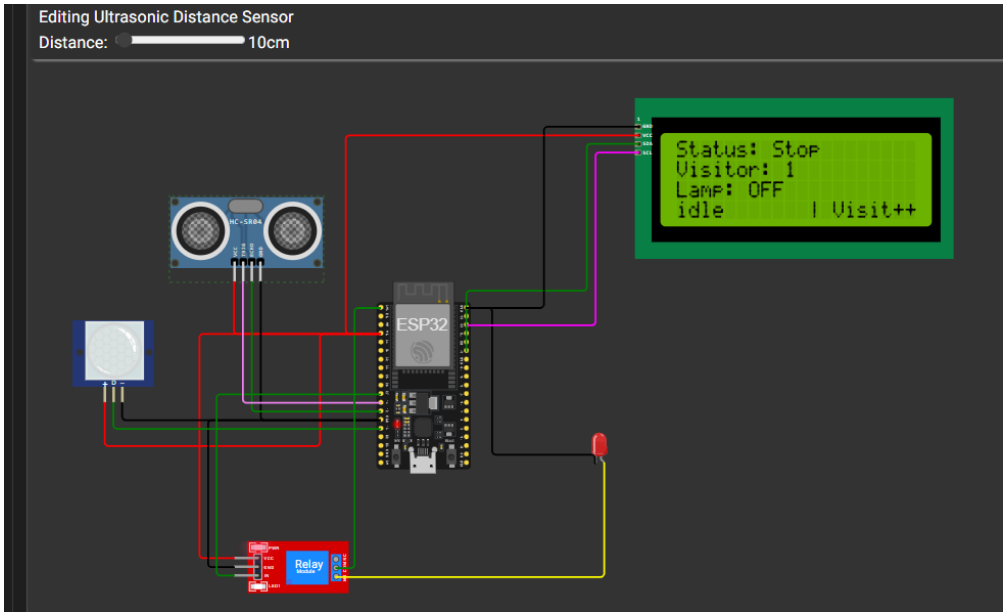


d) Pada tahap **System Active**, setelah login berhasil dan sistem aktif, perangkat melakukan proses koneksi yaitu menghubungkan diri ke jaringan WiFi *Wokwi-GUEST* hingga memperoleh alamat IP 10.10.0.2, kemudian melanjutkan koneksi ke server *blynk.cloud* yang diarahkan ke *sgp1.blynk.cloud:80*, dan setelah koneksi cloud berhasil sistem berjalan penuh dengan pembacaan sensor gerakan dan jarak, memperbarui waktu operasi, serta menyalakan indikator LED secara berkala untuk memastikan status ditampilkan secara real-time.

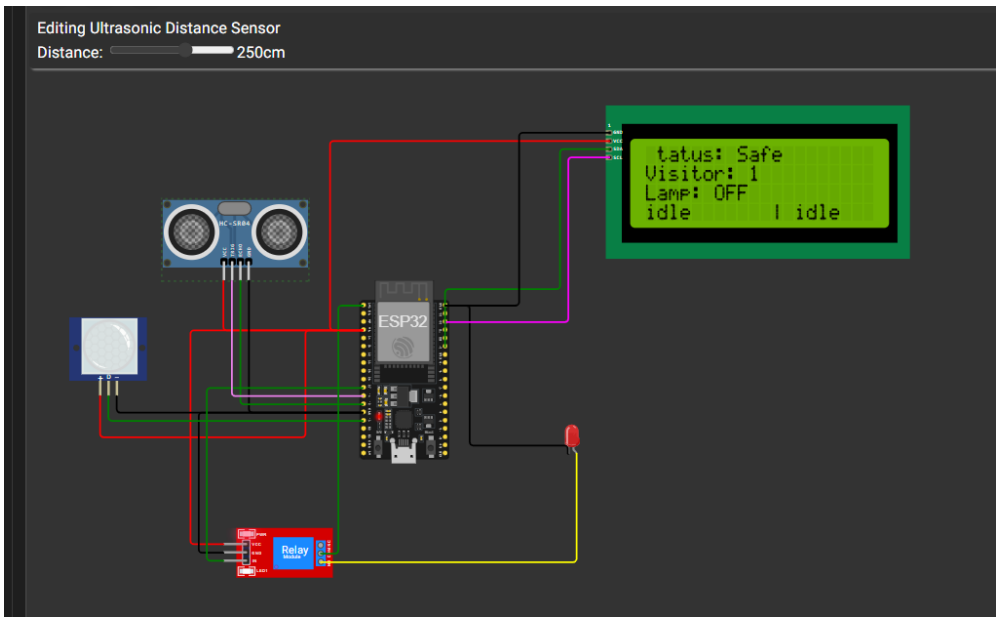



```
Login berhasil! Sistem aktif.
[6202] Connecting to Wokwi-GUEST
[7989] Connected to WiFi
[7990] IP: 10.10.0.2
[7991]
    ___  __          __
   / _ )/ /_ _____  / /__
  / _  / / // / _ \/  '_/
 /____/_/\_, /_//_/_/\_\
        /___/ v1.3.2 on ESP32

 #StandWithUkraine    https://bit.ly/swua


[8001] Connecting to blynk.cloud:80
[8765] Redirecting to sgp1.blynk.cloud:80
[8767] Connecting to sgp1.blynk.cloud:80
[9482] Ready (ping: 81ms).
```

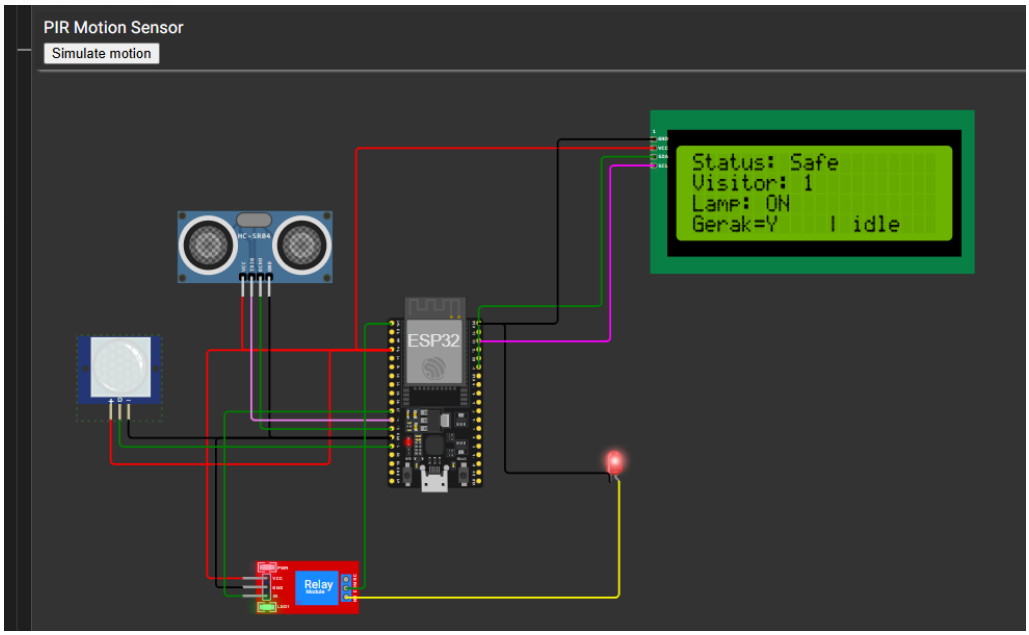
e) Pada tahap **Ultrasonic Sensor Logic**, sistem membaca jarak objek dalam rentang 0–400 cm. Jika jarak yang terdeteksi lebih kecil dari batas maksimum radar yaitu 90 cm, maka penghitung visitor (V4) akan bertambah, status `visitorStatus` berubah menjadi “Stop”, dan Safety Status dikirim ke V9 untuk ditampilkan di dashboard.



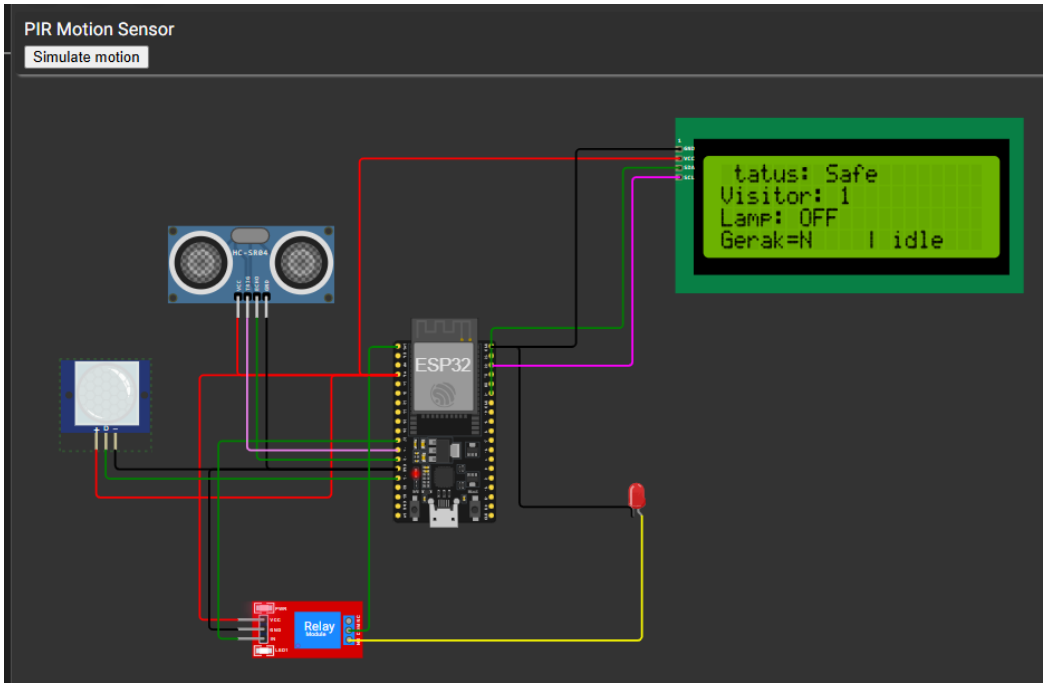
f) Sedangkan jika jarak yang terdeteksi ≥ 90 cm, maka status `visitorStatus` menjadi “Safe” dan Safety Status tetap dikirim ke V9, dengan nilai jarak yang terukur selalu dikirim ke V5 agar dapat dipantau secara real-time oleh pengguna.



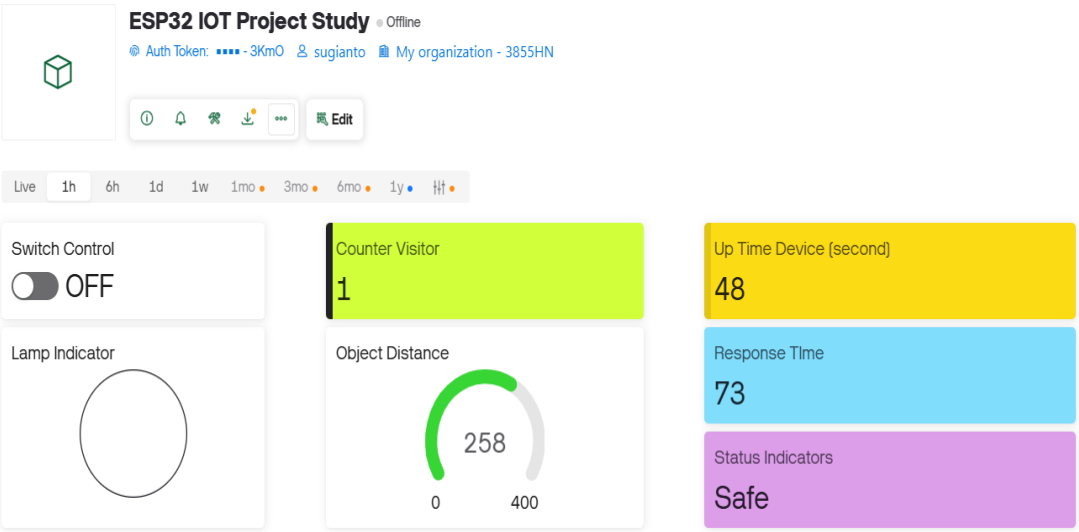
g) Jika PIR mendeteksi gerakan, maka lampu akan menyala dengan relay aktif, status PIR berubah menjadi “Gerak=Y”, dan safety status tetap “Safe” kecuali sensor ultrasonik mendeteksi adanya visitor.



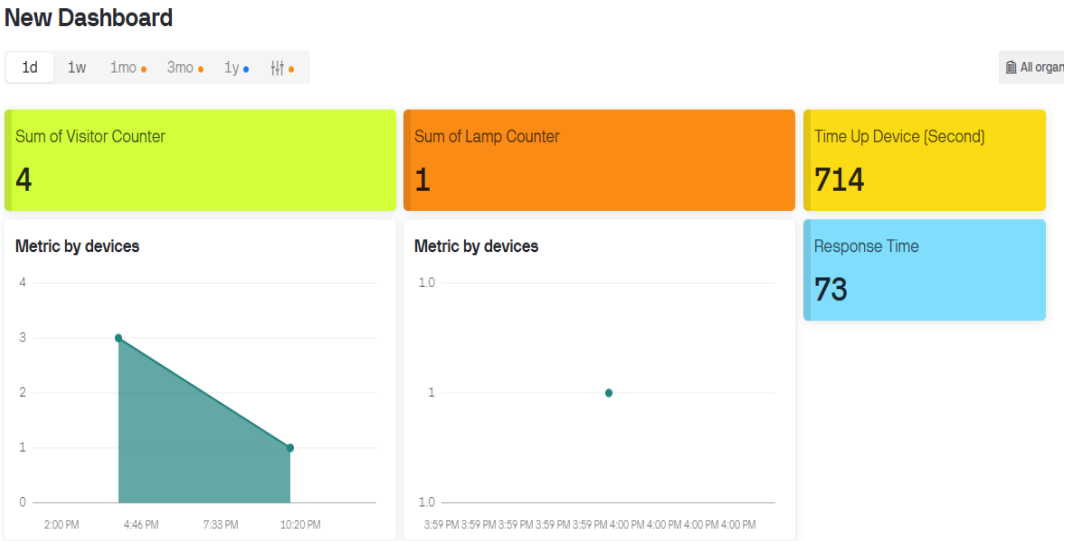
h) Jika tidak ada gerakan yang terdeteksi selama waktu tunggu satu detik, maka lampu akan otomatis mati dengan relay nonaktif, status PIR berubah menjadi “Gerak=N”, dan sistem mengirimkan event log ke Blynk untuk dicatat di dashboard.



i) Pada platform **Blynk IoT**, *devices* berfungsi sebagai representasi fisik dari perangkat keras yang terhubung ke sistem, misalnya ESP32 beserta sensor PIR, sensor ultrasonik, dan aktuator seperti relay atau LED. Setiap device memiliki autentikasi berupa *auth token* yang digunakan untuk menghubungkan perangkat ke cloud Blynk melalui jaringan WiFi, sehingga data dari sensor dapat dikirimkan secara real-time dan perintah dari aplikasi dapat diterima untuk mengontrol aktuator. Dengan adanya konsep device ini, Blynk mampu mengelola banyak perangkat sekaligus, masing-masing dengan data streams dan konfigurasi yang terpisah.



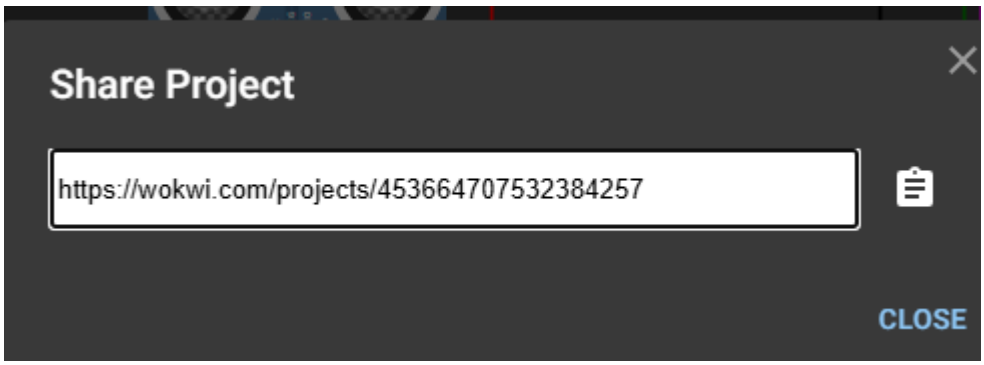
j) Sementara itu, **dashboard Blynk** adalah antarmuka visual yang memungkinkan pengguna memantau status perangkat dan berinteraksi dengan sistem IoT secara langsung. Dashboard menampilkan berbagai widget yang terhubung ke data streams, seperti tombol untuk menyalakan atau mematikan lampu, indikator status keamanan, penghitung visitor, grafik jarak objek, serta informasi uptime perangkat. Melalui dashboard, pengguna tidak hanya dapat melihat data sensor secara real-time, tetapi juga melakukan kontrol dan evaluasi performa sistem dengan cara yang lebih mudah, interaktif, dan terpusat.



Lampiran

Lampiran A – WorkWi simulation file (.wif)

https://wokwi.com/projects/453664707532384257



Lampiran B – Coding WorkWi (sketch.ino)

```
#define BLYNK_TEMPLATE_ID "TMPL6ClFgS-2L"
#define BLYNK_TEMPLATE_NAME "ESP32 IOT Project Study"
#define BLYNK_AUTH_TOKEN "JC6k0TRdHYwcTjCiOKKMBStae2eF3KmO"
#define BLYNK_PRINT Serial

#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <NewPing.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// --- Konfigurasi Authentication by Serial Monitor ---
String inputCode = "";
String correctCode = "1234";
bool isLoggedIn = false;

// --- Konfigurasi WiFi ---
char ssid[] = "Wokwi-GUEST";
char pass[] = "";

// --- Definisi Pin Hardware ---
#define LED_BUILTIN_PIN 2
#define PIR_PIN 13
#define RELAY_PIN 27
```

```
#define TRIG_PIN 14
#define ECHO_PIN 12
#define MAX_DISTANCE 400
NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);
BlynkTimer timer;
LiquidCrystal_I2C lcdBig(0x27, 20, 4);

// --- Deklarasi Variabel ---
bool systemStarted = false;
int counterOrang = 0;
unsigned long lastMotionTime = 0;
bool motionDetected = false;
bool lastSonarState = false;
unsigned int maxDistanceRadar = 90;
String pirMsg = "idle";
String sonarMsg = "idle";
String visitorStatus = "Safe";
const unsigned long TIMEOUT_LAMP_OFF = 1000; // 1 detik
unsigned long pirTriggerTime = 0;


void setup() {
  Serial.begin(115200);
  Serial.println("Masukkan Kode untuk mengaktifkan sistem:");
  fn_lcdBig_Initialization();
  fn_PinHardware_Initialization();
  fn_turnOffLamp();
}

void loop() {
  if (!isLoggedIn && Serial.available()) {
    inputCode = Serial.readStringUntil('\n');
    inputCode.trim();
    if (inputCode == correctCode) {
      isLoggedIn = true;
      fn_RunonLoginSuccess();
      fn_startSystem();
    } else {
      Serial.println("Kode salah. Coba lagi.");
    }
  }

  if (isLoggedIn) {
    Blynk.run();
    timer.run();
  }
}
```

```
void fn_checkPIR() {
  int pirState = digitalRead(PIR_PIN);

  if (pirState == HIGH && !motionDetected) {
    pirTriggerTime = millis(); // catat waktu trigger
    fn_turnOnLamp();
    unsigned long responseTime = millis() - pirTriggerTime;
    fn_PrintMessageToConsole("Response PIR->Lamp: " + String(responseTime) + " ms");
    Blynk.virtualWrite(V10, responseTime); // kirim ke V10
    pirMsg = "Gerak=Y";
  }

  if (pirState == LOW && motionDetected) {
    if (millis() - lastMotionTime >= TIMEOUT_LAMP_OFF) {
      fn_turnOffLamp();
      pirMsg = "Gerak=N";
      fn_PrintMessageToConsole("Tidak ada gerakan, Lampu OFF");
    }
  }
  fn_refreshLCDBig();
}


void fn_checkUltrasonic() {
  unsigned int distance = sonar.ping_cm();
  fn_PrintMessageToConsole(String("distance ") + distance);
  Blynk.virtualWrite(V5, distance); // kirim jarak ke V5

  if (distance > 0 && distance < maxDistanceRadar) {
    if (!lastSonarState) {
      lastSonarState = true;
            counterOrang++;
      sonarMsg = "Visit++";
      visitorStatus = "Stop";
      fn_PrintMessageToConsole("Visitor bertambah");
      Blynk.virtualWrite(V4, counterOrang);
      Blynk.virtualWrite(V9, visitorStatus);
    }
  } else {
    lastSonarState = false;
    sonarMsg = "idle";
    visitorStatus = "Safe";
    fn_PrintMessageToConsole("Tidak ada visitor baru");
    Blynk.virtualWrite(V9, visitorStatus);
  }
  fn_refreshLCDBig();
}
```

```
void fn_turnOnLamp() {
  digitalWrite(RELAY_PIN, HIGH);
  motionDetected = true;
  lastMotionTime = millis();
  Blynk.virtualWrite(V0, 1);
  Blynk.virtualWrite(V1, 1);
  fn_PrintMessageToConsole("Lampu ON");
}


void fn_turnOffLamp() {
  digitalWrite(RELAY_PIN, LOW);
  motionDetected = false;
  Blynk.virtualWrite(V0, 0);
  Blynk.virtualWrite(V1, 0);
  fn_PrintMessageToConsole("Lampu OFF");
}


void fn_refreshLCDBig() {
  // Baris 1: Status visitor Stop/Safe
  lcdBig.setCursor(0, 0);
  lcdBig.print("Status: ");
  lcdBig.print(visitorStatus);
  lcdBig.print("        ");

  // Baris 2: Visitor count
  lcdBig.setCursor(0, 1);
  lcdBig.print("Visitor: ");
  lcdBig.print(counterOrang);
  lcdBig.print("        ");

  // Baris 3: Lamp ON/OFF
  lcdBig.setCursor(0, 2);
  lcdBig.print("Lamp: ");
  lcdBig.print(motionDetected ? "ON " : "OFF");
  lcdBig.print("        ");

  // Baris 4: Gabungan pesan PIR dan Sonar
  lcdBig.setCursor(0, 3);
  lcdBig.print("                    ");
  lcdBig.setCursor(0, 3);
  lcdBig.print(pirMsg);
  lcdBig.setCursor(10, 3);
  lcdBig.print(" | ");
  lcdBig.print(sonarMsg);
```

```
}


void fn_lcdBig_Initialization() {
  lcdBig.init();
  lcdBig.backlight();
  lcdBig.setCursor(0, 0);
  lcdBig.print("System Locked...");
  lcdBig.setCursor(0, 1);
  lcdBig.print("Login Required");
}

void fn_PinHardware_Initialization() {
  pinMode(PIR_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);
  pinMode(LED_BUILTIN_PIN, OUTPUT);
}

void fn_RunonLoginSuccess() {
  Serial.println("Login berhasil! Sistem aktif.");
  lcdBig.clear();
  lcdBig.setCursor(0,0);
  lcdBig.print("System Active");
  lcdBig.setCursor(0,1);
  lcdBig.print("Welcome!");
}

void fn_startSystem() {
  if (!systemStarted) {
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
    timer.setInterval(150L, fn_checkUltrasonic);
    timer.setInterval(500L, fn_checkPIR);
    timer.setInterval(1000L, fn_myTimerEvent);
    timer.setInterval(1000L, fn_Blink_LED);
    systemStarted = true;
  }
}


void fn_PrintMessageToConsole(String iMsg) {
  fn_PrintCurrentTime();
  Serial.println(iMsg);
}


void fn_PrintCurrentTime() {
  unsigned long currentTime = millis();
  unsigned long seconds = currentTime / 1000;
```

```
  unsigned long minutes = seconds / 60;
  unsigned long hours = minutes / 60;
  seconds %= 60;
  minutes %= 60;
  hours %= 24;
  Serial.print("[");
  fn_padZero(hours);
  Serial.print(":");
  fn_padZero(minutes);
  Serial.print(":");
  fn_padZero(seconds);
  Serial.print("]: ");
}


void fn_padZero(unsigned long value) {
  if (value < 10) Serial.print('0');
  Serial.print(value);
}


void fn_Blink_LED() {
  static bool state = false;
  state = !state;
  digitalWrite(LED_BUILTIN_PIN, state ? HIGH : LOW);
}


void fn_myTimerEvent() {
  Blynk.virtualWrite(V2, millis() / 1000);
}


BLYNK_WRITE(V0) {
  int buttonState = param.asInt();
  if (buttonState == 1) fn_turnOnLamp();
  else fn_turnOffLamp();
}

BLYNK_CONNECTED() {
  Blynk.setProperty(V3, "offImageUrl","https://imgur.com/WHx1WG6");
  Blynk.setProperty(V3, "onImageUrl","https://imgur.com/a/gASrmP1");
}
```

Lampiran C – Diagram WorkWi (diagram.json)

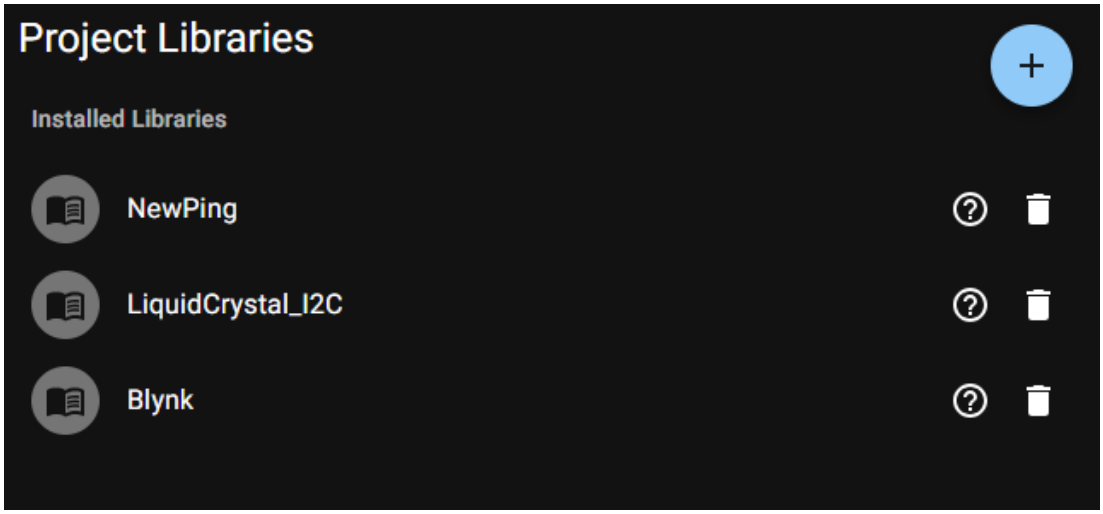

```
{
  "version": 1,
  "author": "Anonymous maker",
  "editor": "wokwi",
  "parts": [
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": -9.6, "left": 129.64, "attrs": {} },
    { "type": "wokwi-pir-motion-sensor", "id": "pir1", "top": 32.8, "left": -208.98, "attrs": {} },
    { "type": "wokwi-relay-module", "id": "relay1", "top": 278.6, "left": -28.8, "attrs": {} },
    {
      "type": "wokwi-hc-sr04",
      "id": "ultrasonic1",
      "top": -104.1,
      "left": -100.1,
      "attrs": { "distance": "20" }
    },
    { "type": "wokwi-led", "id": "led1", "top": 150, "left": 359, "attrs": { "color": "red" } },
    {
      "type": "wokwi-lcd2004",
      "id": "lcd2",
      "top": -214.4,
      "left": 418.4,
      "attrs": { "pins": "i2c", "addr": "0x26" }
    }
  ],
  "connections": [
    [ "esp:TX", "$serialMonitor:RX", "", [] ],
    [ "esp:RX", "$serialMonitor:TX", "", [] ],
    [ "ultrasonic1:VCC", "esp:VN", "red", [ "v0" ] ],
    [ "pir1:VCC", "esp:VN", "red", [ "v48", "h240", "v-124.8", "h-19.2" ] ],
    [ "relay1:VCC", "esp:VN", "red", [ "h-38.4", "v-249.6", "h57.6" ] ],
    [ "pir1:GND", "esp:GND.1", "black", [ "v0" ] ],
    [ "relay1:GND", "esp:GND.1", "black", [ "h-28.8", "v-163.6", "h9.6" ] ],
    [ "ultrasonic1:GND", "esp:GND.1", "black", [ "v153.6", "h-10.8" ] ],
    [ "pir1:OUT", "esp:13", "green", [ "v0" ] ],
    [ "relay1:IN", "esp:27", "green", [ "h-19.2", "v-0.2" ] ],
    [ "ultrasonic1:ECHO", "esp:12", "green", [ "v0" ] ],
    [ "ultrasonic1:TRIG", "esp:14", "violet", [ "v0" ] ],
    [ "led1:A", "relay1:NO", "yellow", [ "v0" ] ],
    [ "relay1:COM", "esp:3V3", "green", [ "h20.4", "v-289.4", "h9.6" ] ],
    [ "led1:C", "esp:GND.2", "black", [ "v-9.6", "h-114.8", "v-163.2" ] ],
    [ "lcd2:SCL", "esp:22", "magenta", [ "h-48", "v125.1" ] ],
    [ "lcd2:SDA", "esp:21", "green", [ "h-57.6", "v163.4", "h-9.6" ] ],
    [ "lcd2:VCC", "esp:VN", "red", [ "h-326.4", "v115.3" ] ],
    [ "lcd2:GND", "esp:GND.2", "black", [ "h-105.6", "v96" ] ]
  ],
  "dependencies": {}
```

}

Lampiran D – Library WorkWi (libraries.txt)

```
# Wokwi Library List
# See https://docs.wokwi.com/guides/libraries

NewPing
LiquidCrystal_I2C
Blynk
```

Lampiran E – Library Manager WorkWi



Lampiran F – Platform Blynk

Lampiran G – Blynk Auth Token

```
#define BLYNK_TEMPLATE_ID "TMPL6ClFgS-2L"
#define BLYNK_TEMPLATE_NAME "ESP32 IOT Project Study"
#define BLYNK_AUTH_TOKEN "JC6k0TRdHYwcTjCiOKKMBStae2eF3KmO"
```

Lampiran H – Blynk Data Streams

Datastreams

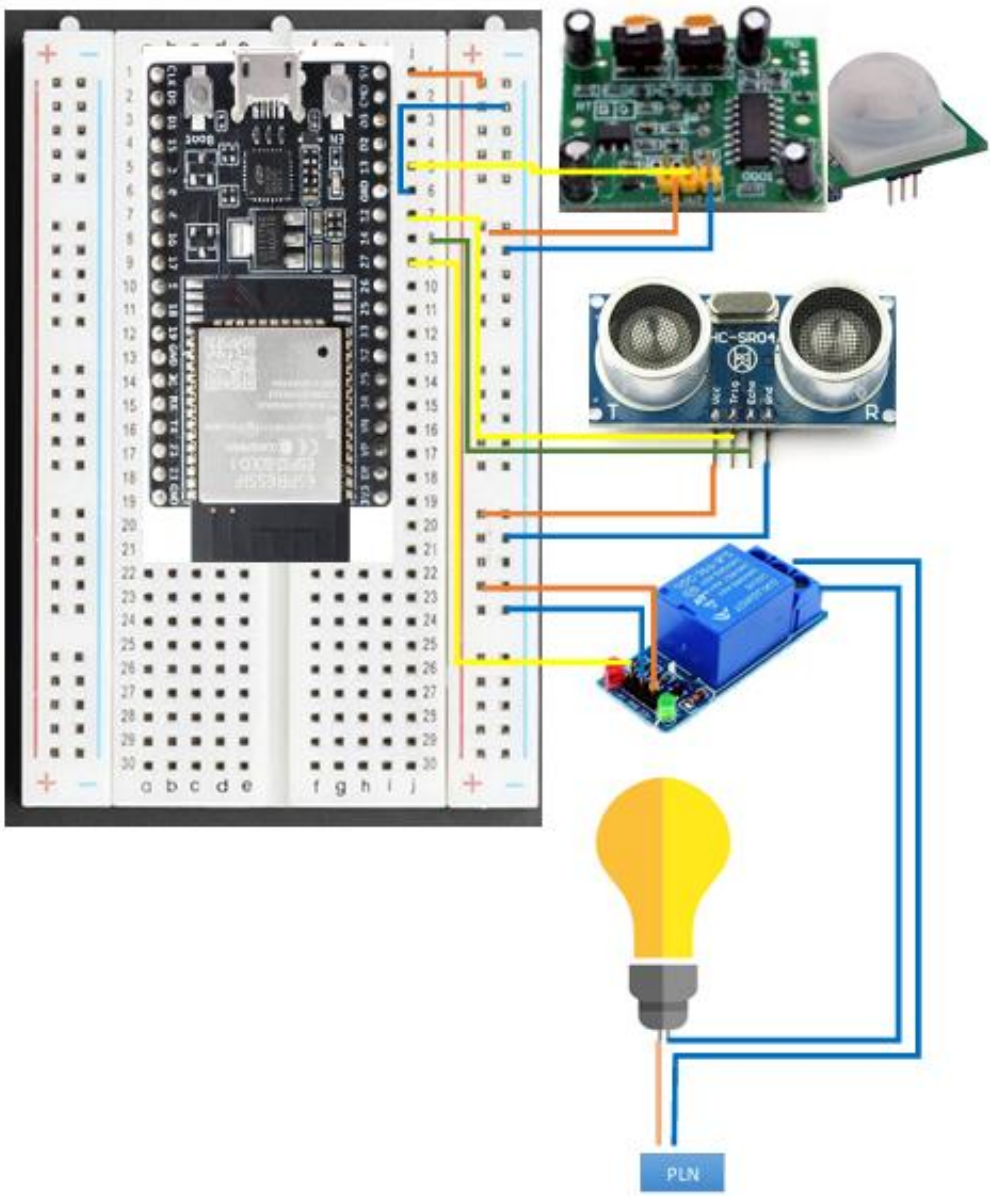
Search datastream

+ New Datastream

11 Datastreams

ID	Name	Pin	Color	Data Type	Units	Is Raw	Min	Max	Decimals	Default Value	Automation Type	Actions
1	Lamp Control	V0		Integer		false	0	1	--	0	Switch	
2	Lamp Value	V1		Integer		false	0	1	--	0	Switch	
3	Time Up	V2		Integer		false	0	99999999	--	0	Switch	
4	Button Image	V3		String		false			--		Color	
5	Visitor Counter	V4		Integer		false	0	9999999	--	0	Switch	
6	Object Distance	V5		Integer		false	0	400	--	0	Switch	
7	Fan Control	V6		Integer		false	0	1	--	0	Switch	
8	Fan Value	V7		Integer		false	0	1	--	0	Switch	
9	Temperature Value	V8		Double		false	0	255	#.##		Sensor	
10	Status Indicators	V9		String		false			--		Color	
11	Response Time	V10		Double		false	0	99999999	#.##		Sensor	

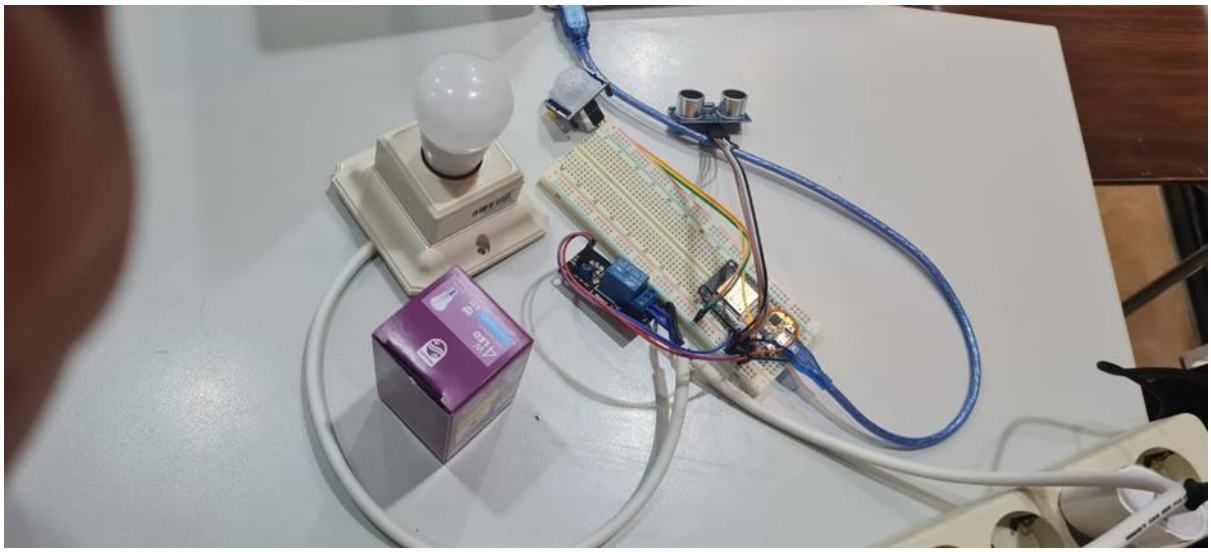
Lampiran I – Diagram Blok Sistem (Perangkat Fisik)

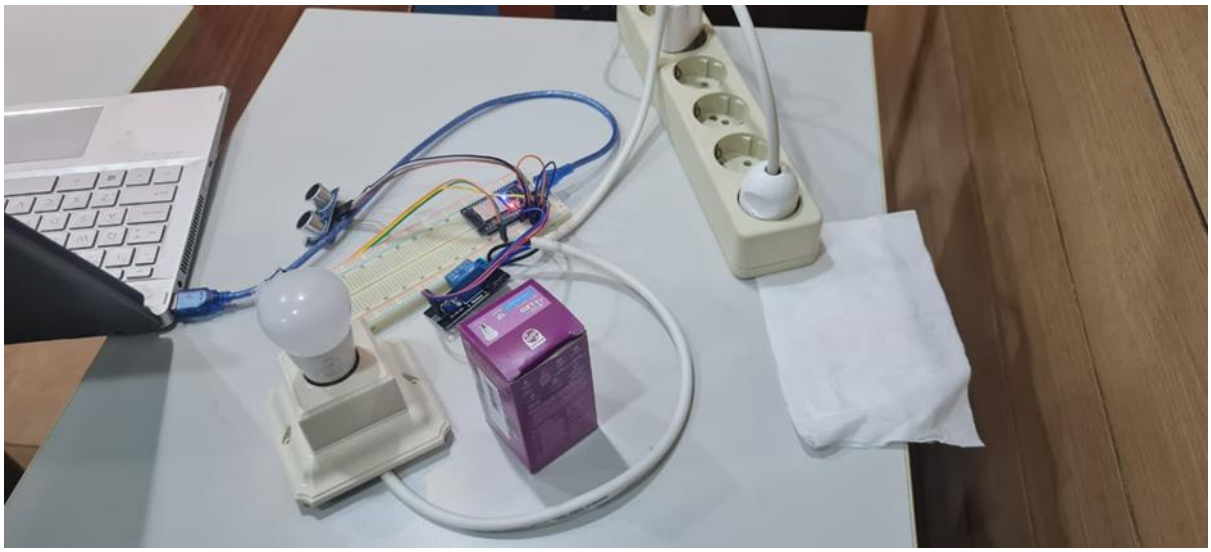


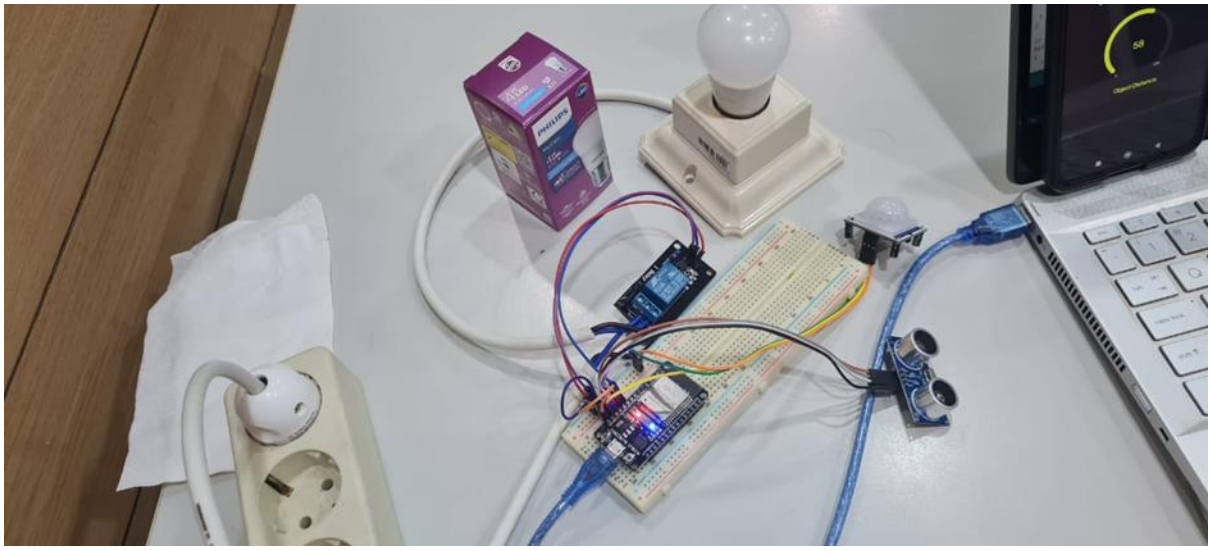
Gambar A.1 Diagram Blok Sistem IoT

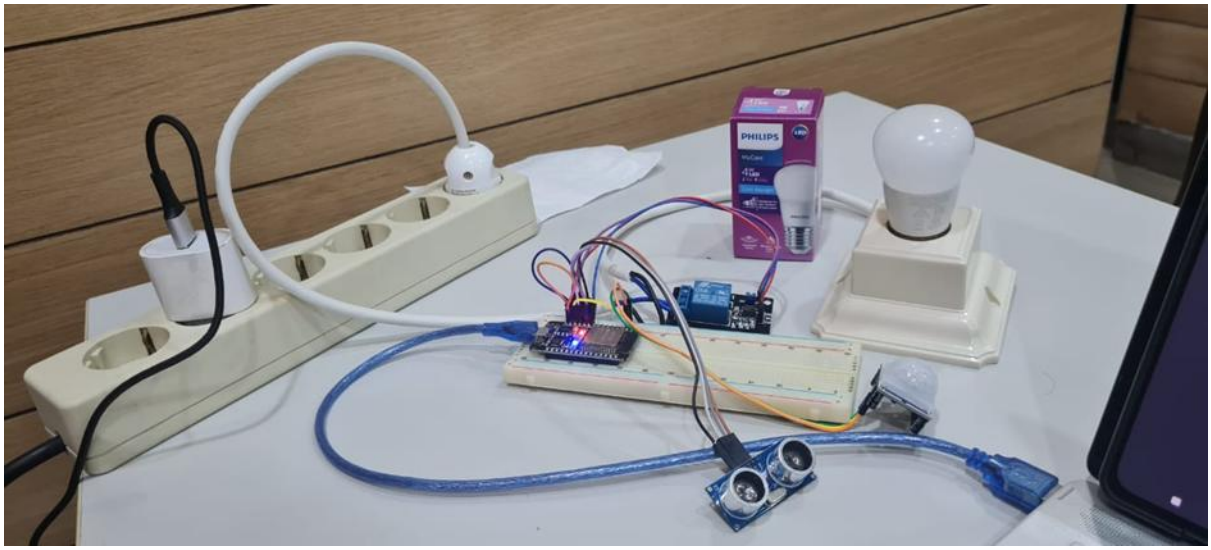
Relay	PIR Motion	HC-SR042	ESP32	
VCC	VCC	VCC	VN	3V3
GND	GND	GND	GND	GND
	Out		D13	D15
		Echo	D12	D2
		Trig	D14	D4
In			D27	RX2
			D26	TX2
			D25	D05
			D33	D18
			D32	D19
			D35	D21
			D34	RX0
			VN	TX0
			VP	D22
			EN	D23

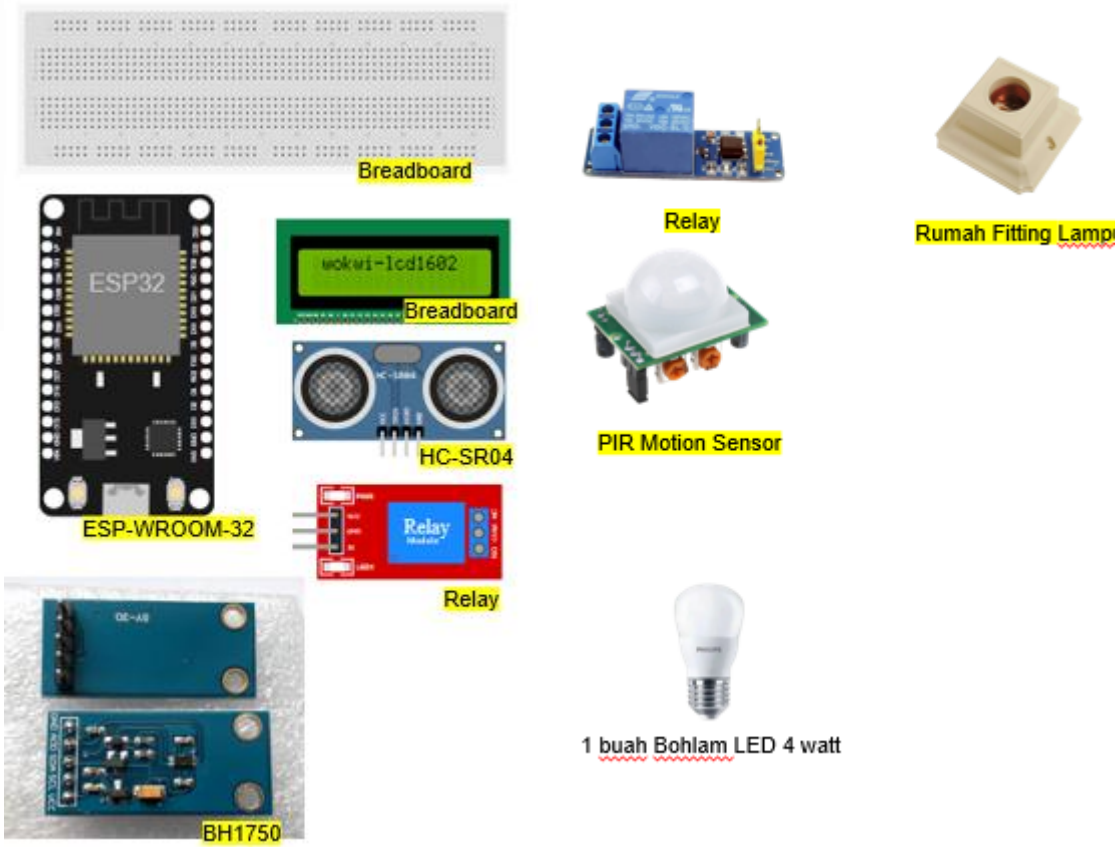
Lampiran J – Dokumentasi Mock-up Perangkat Fisik dari beberapa angel







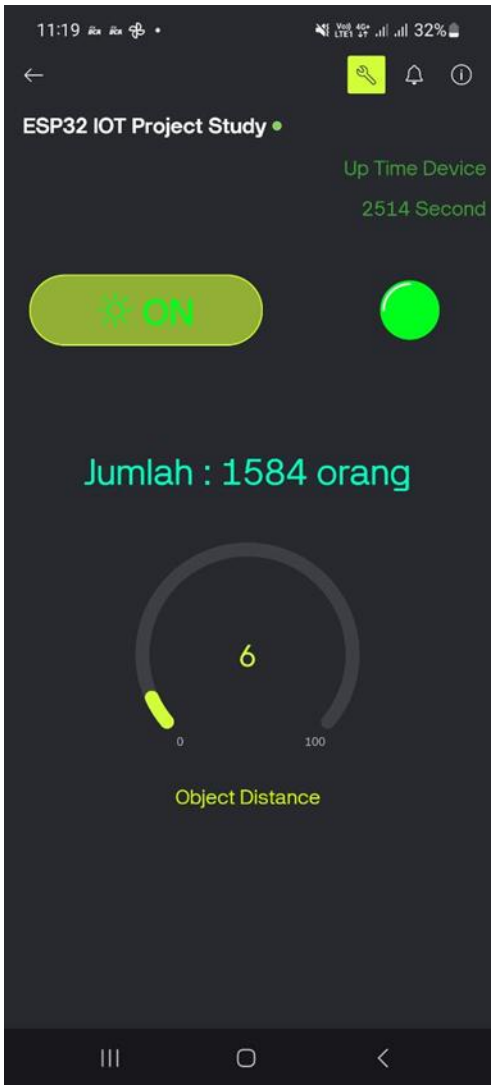

PHILIPS



Gambar C.2 Detail Komponen Mock-up

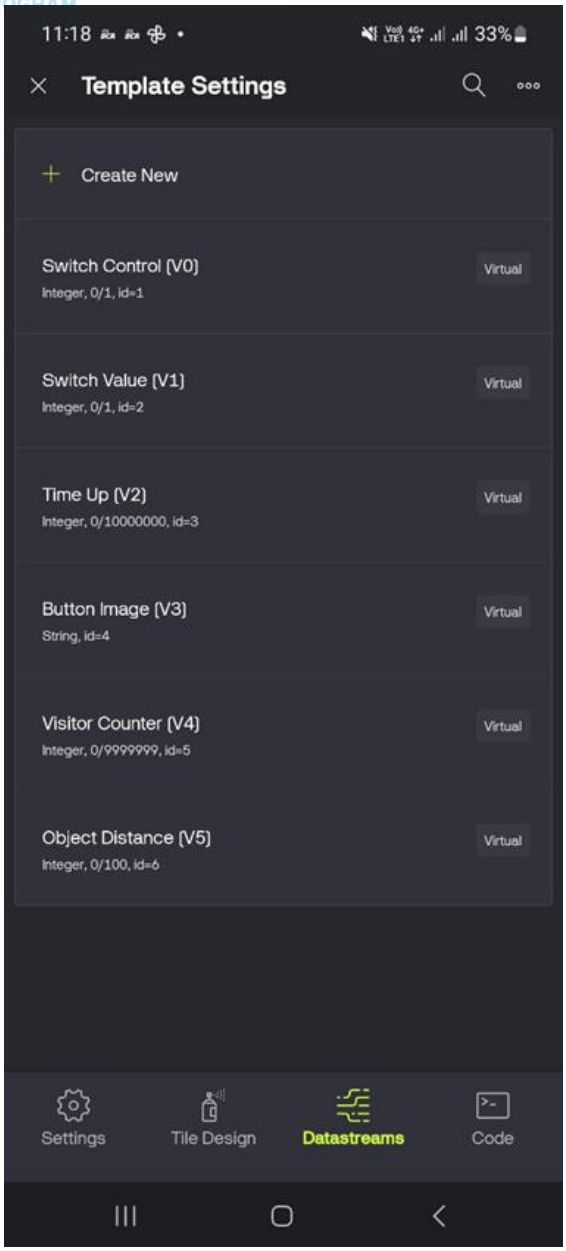
Lampiran K – Dokumentasi Mock-up Dashboard Aplikasi

Gambar Dashboard dengan Lampu Off

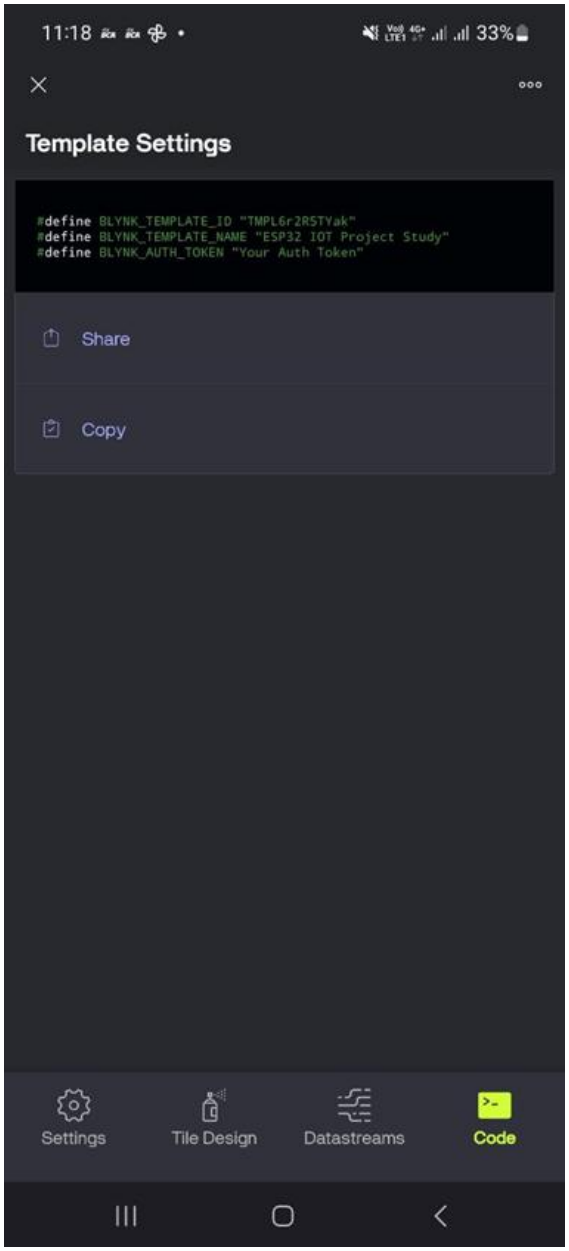


Gambar Dashboard dengan Lampu On

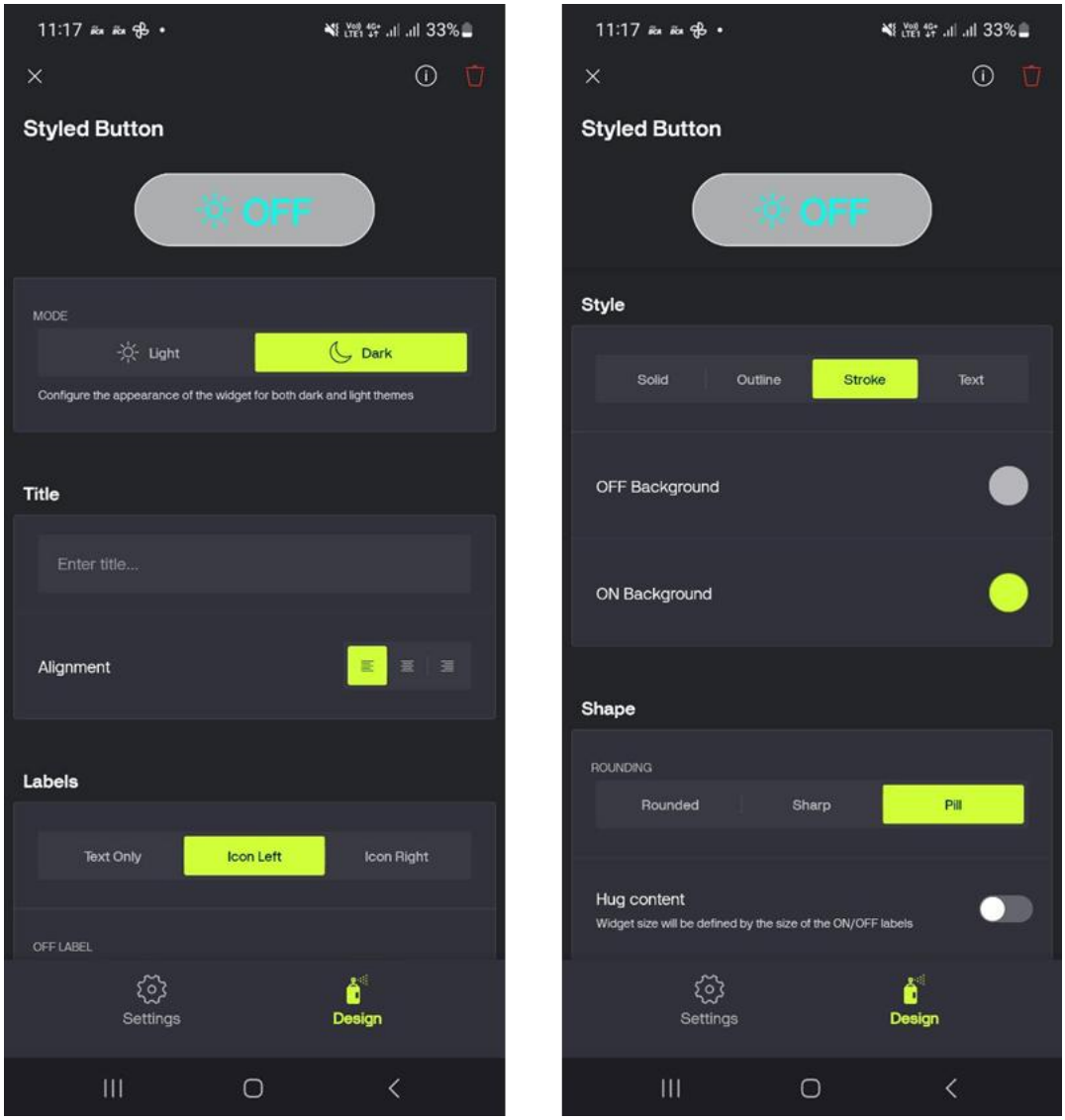
BINUS UNIVERSITY GRADUATE PROGRAM
People Innovation Excellence



Gambar Dashboard Data Stream



Gambar Dashboard Code



Gambar Dashboard Design Style

Lampiran L – Dokumentasi Video Demonstrasi Mock-up Perangkat Fisik

Video menampilkan simulasi kerja sensor deteksi pergerakan untuk menyalakan dan mematikan listrik. Kondisi listrik divisualisasikan menggunakan lampu bohlam dan dikendalikan secara otomatis maupun manual melalui perangkat HP.

Durasi Video

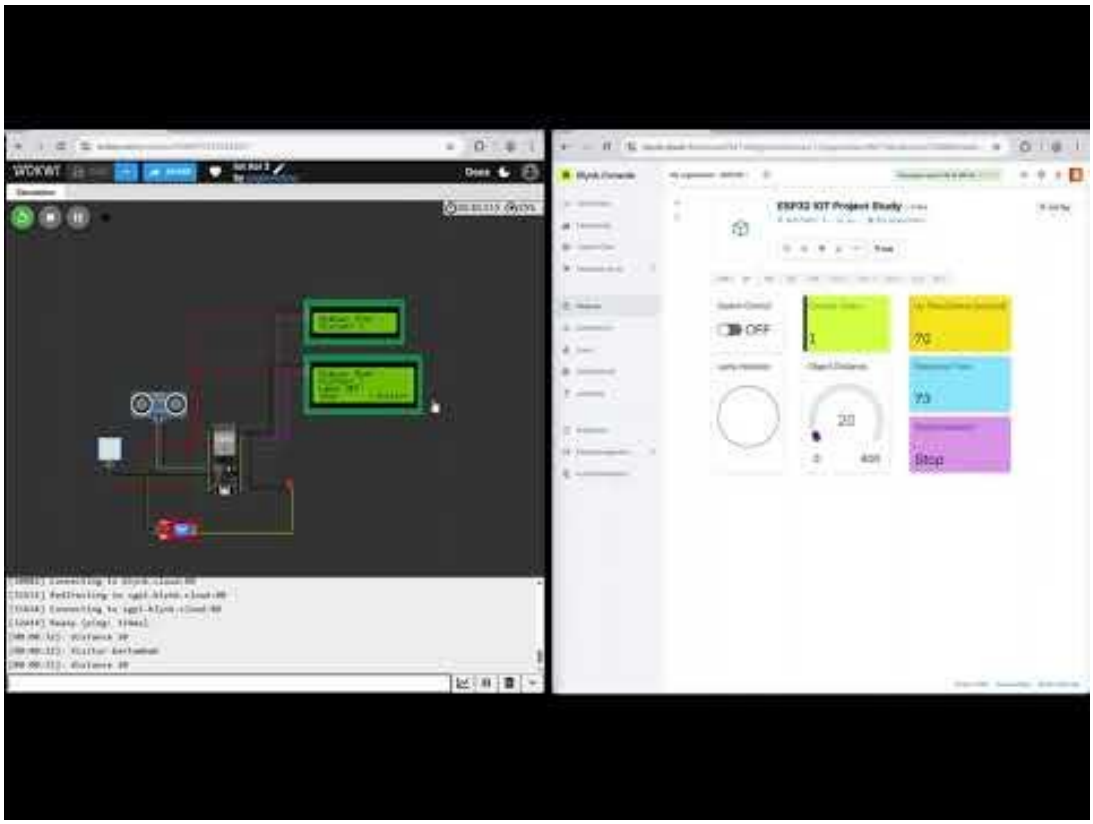
± 2 menit 24 detik

Akses Video

Link Youtube:

https://youtube.com/shorts/0UEsfs9zyao?feature=share

Lampiran M – Dokumentasi Video Demonstrasi Mock-up Perangkat Virtual WorkWI dan Blynk



Durasi Video

± 1 menit 55 detik

Akses Video

Link:

https://youtu.be/Uxs0q0tw4q0